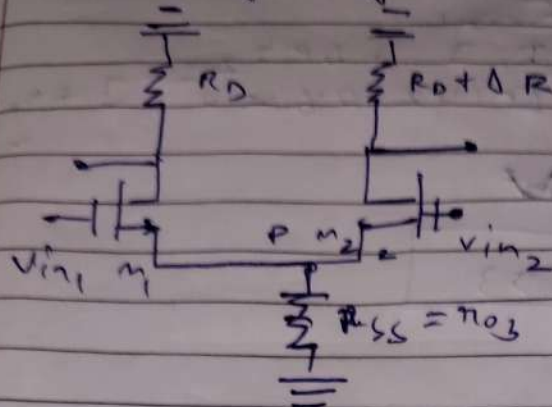


small-signal gain :-



$$i_1 + i_2 = \frac{v_p}{R_{SS}}$$

$$i_1 = g_{m1} (v_{in1} - v_p)$$

$$i_2 = g_{m2} (v_{in2} - v_p)$$

$$v_p = \frac{g_{m1} v_{in1} + g_{m2} v_{in2}}{g_{m1} + g_{m2} + \frac{1}{R_{SS}}}$$

$$v_p = \frac{141.4 \times 10^{-6} \times 0.5 + 142.8 \times 10^{-6} \times 0.5}{141.4 \times 10^{-6} + 142.8 \times 10^{-6} + \frac{1}{923 \times 10^3}}$$

$$= \frac{70.7 - 71.4}{284.2 + 1.08}$$

$$= - \frac{0.7}{285.3} \approx -2.4 \text{ mV}$$

$$i_1 = 141.4 \times 10^{-6} [-0.5 + 2.4 \times 10^{-3}]$$

$$= 141.4 \times 10^{-6} [-0.5024]$$

$$\approx -71.03 \mu\text{A}$$

$$i_2 = 142.8 \times 10^{-6} [-0.5 + 2.4 \times 10^{-3}]$$

$$= 142.8 \times 10^{-6} [-0.4976]$$

$$\approx -71.05 \mu\text{A}$$

~~$$v_{out1} = v_p = 1.1 \times 10^{-6} \times 0.032 \times 10^5$$~~

$$v_{out1} = -I_1 R_D$$

$$v_{out2} = -I_2 (R_D + \Delta R)$$

$$V_u - V_y = -I_1 R_D + I_1 (R_D + \Delta R)$$

$$V_u - V_y = R_D [I_2 - I_1] + \Delta R I_2$$

$$= 3.175 \times 10^5 [-142.04 \times 10^{-6}] + 6.032$$

$$V_u - V_y = -45.33$$

$$A_{dm} = -45.33$$